Microsoft 365

MD-100 Windows Client

Module 08 : Configure threat protection

Module Agenda

- Explore malware and threat protection
- Explore Microsoft Defender
- Explore device encryption features
- Explore connection security rules
- Explore advanced protection methods

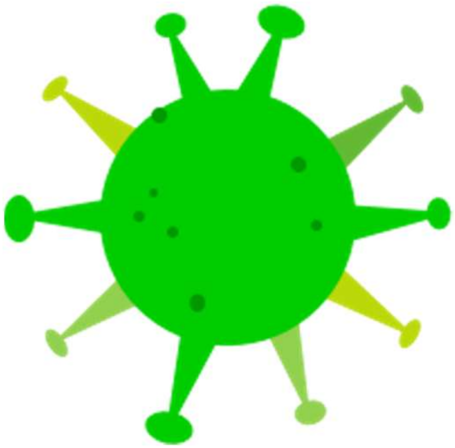
Lesson 1: Explore malware and threat protection

Lesson 1: Explore malware and threat protection

- Explain malware
- Determine possible mitigations for malware threats
- Understand phishing scams
- Explain common network-related security threats
- Determine mitigations for network-related security threats

Explain malware

- Forms of malware include the following:
- Computer viruses
- Computer worms
- Trojan horses
- Ransomware
- Spyware



Determine possible mitigations for malware threats

- Ensure that you apply all software and operating system updates to your devices.
- Ensure that you install and activate anti-malware software on your devices.
- Ensure that anti-malware definitions are current.
- Avoid risky behavior, such as consuming pirated software or media.
- Avoid opening suspicious email attachments, even if they are from senders that you trust.

Understand phishing scams

- Common tricks that cybercriminals use:
- Fake websites
- Fake threats or warnings
- Spoofing companies or people you know

Explain common network-related security threats

- Eavesdropping
- Denial of service (DoS) attack
- Port scanning
- Man-in-the-middle (MITM) attack

Determine mitigations for network-related security threats

It is important to implement a comprehensive approach to network security to ensure that one loophole or omission does not result in another

Attack	Mitigations
Eavesdropping	IPsec, VPNs, DirectAccess, intrusion detection
DoS	Firewalls, perimeter networks, IPsec, server hardening
Port scanning	Server hardening, firewalls
MITM	IPsec, DNSSEC
Virus, malicious code	Software updates

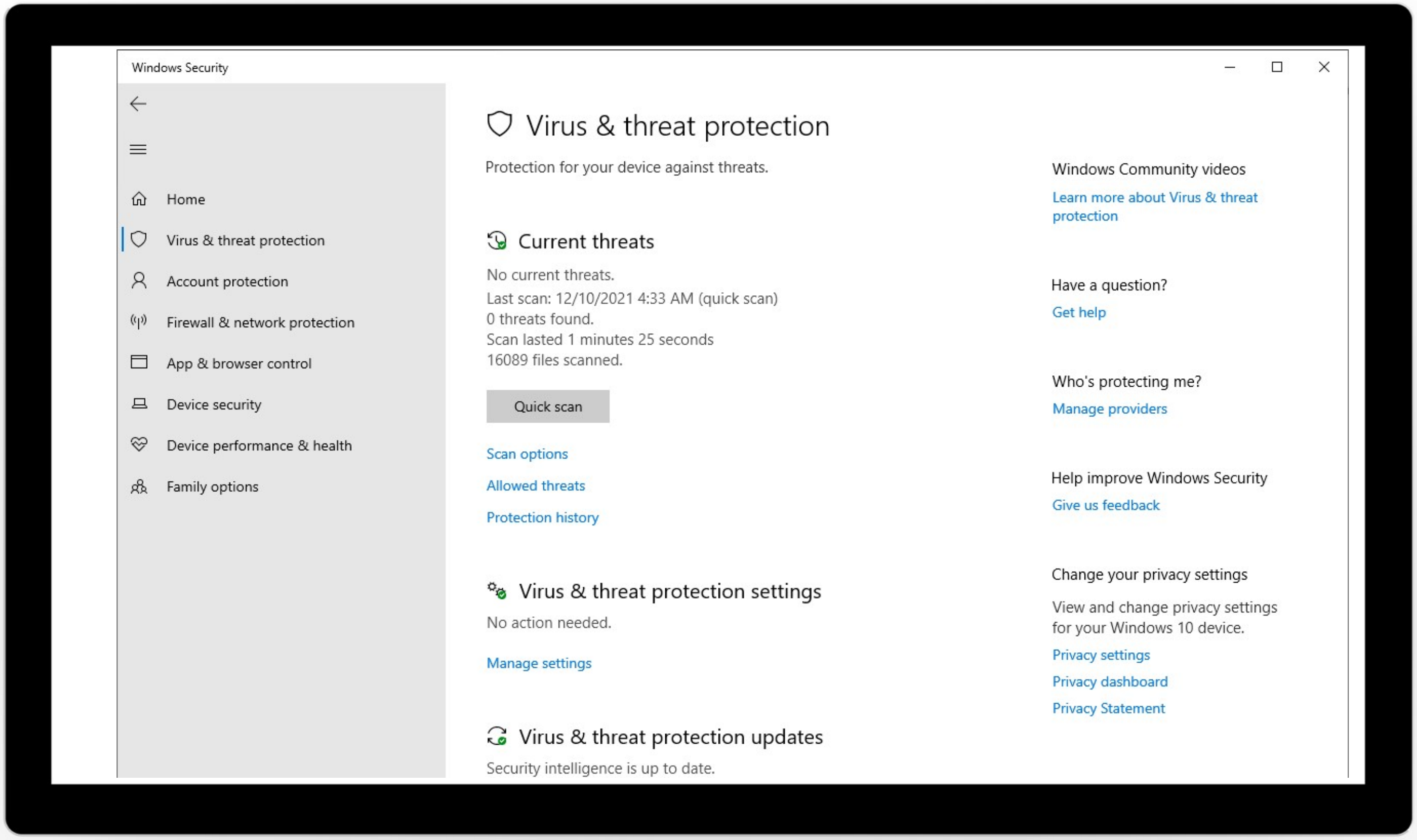
Knowledge Check

Lesson 2: Explore Microsoft Defender

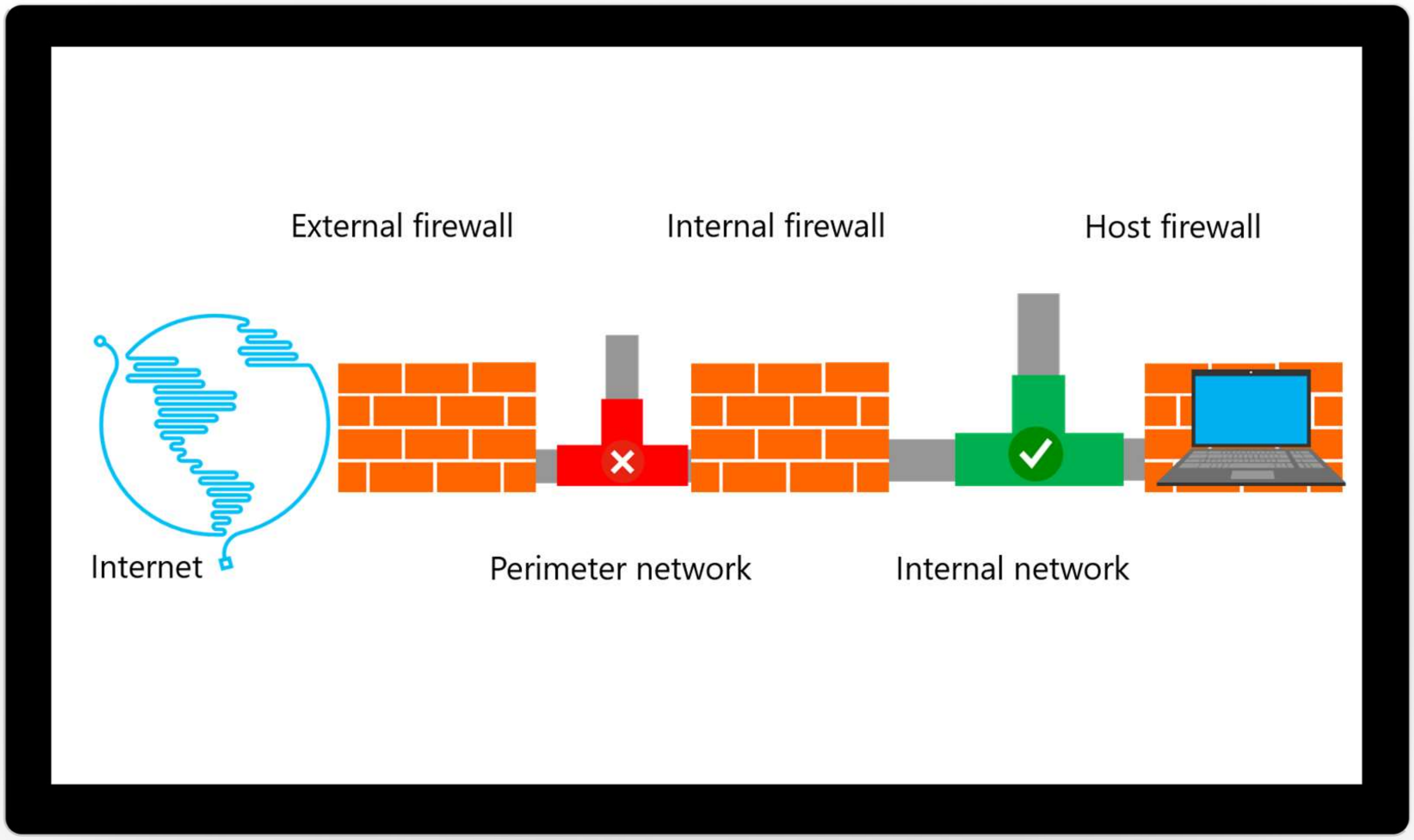
Lesson 2: Explore Microsoft Defender

- Explain Microsoft Defender Antivirus
- Understand Firewalls
- Explain Windows Defender Firewall
- Examine network location profiles
- Explain Windows Defender Firewall with Advanced Security

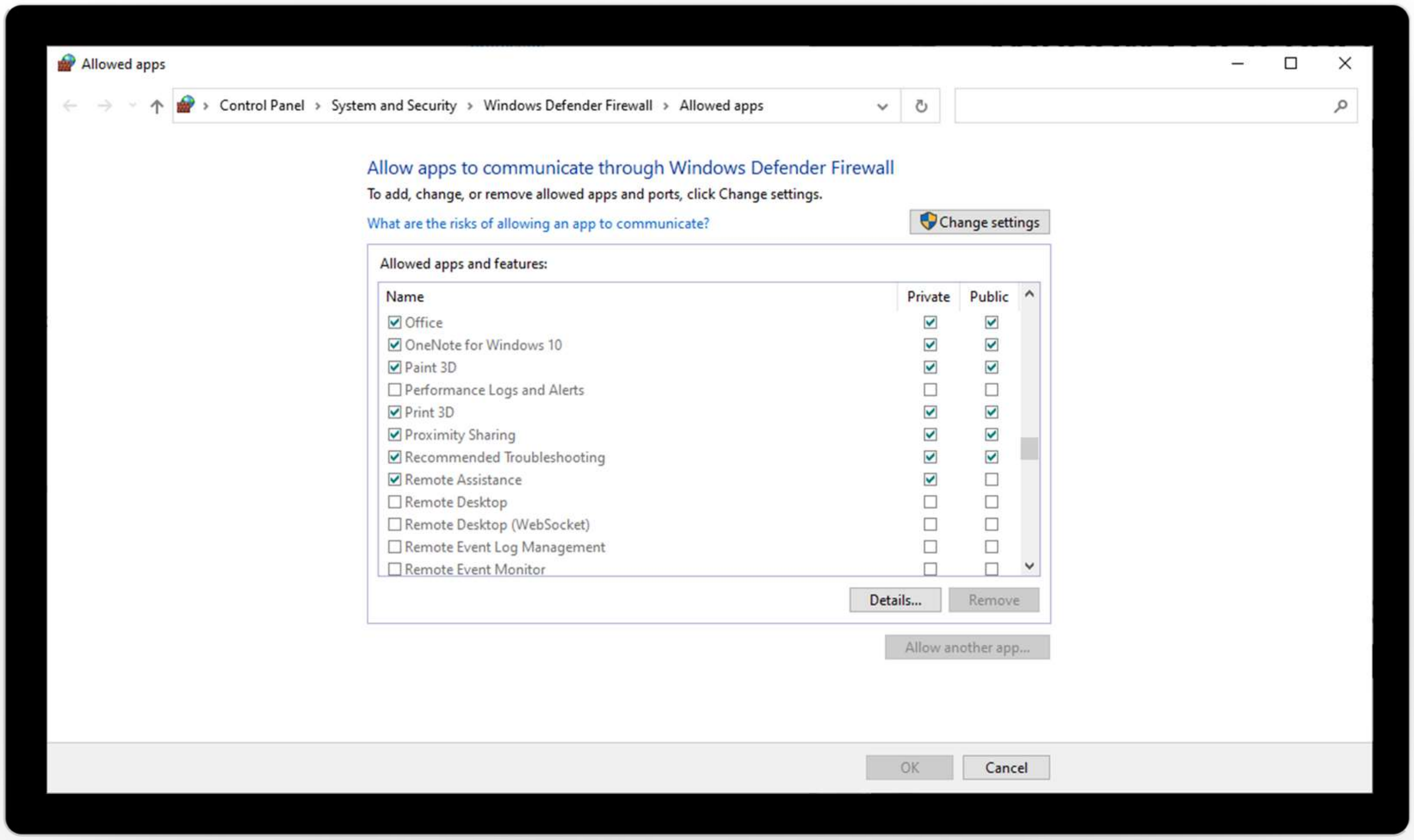
Explain Microsoft Defender Antivirus



Understand Firewalls



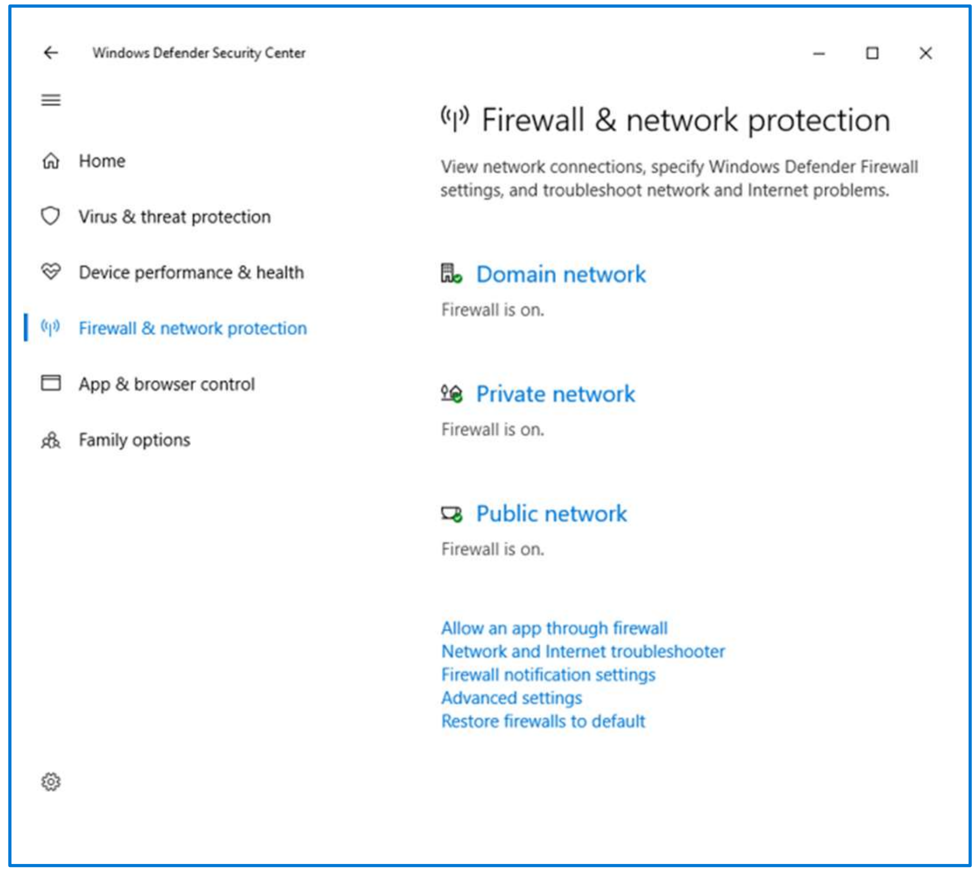
Explain Windows Defender Firewall



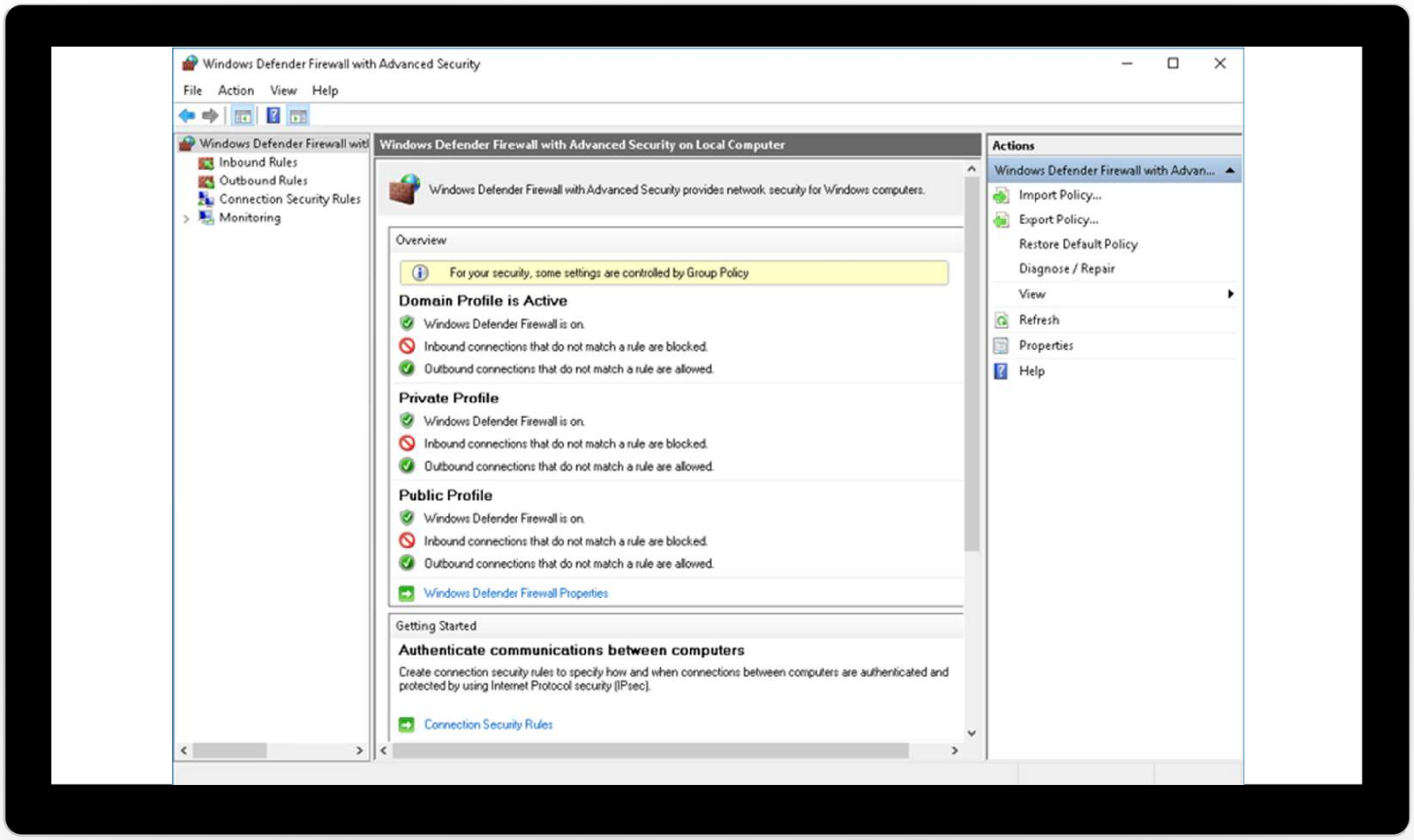
Examine network location profiles

- Windows uses network location awareness to identify connected networks uniquely

- Networks can be classified as one of three network location types:
  - Domain
  - Public
  - Private



Explain Windows Defender Firewall with Advanced Security



Knowledge Check

Lesson 3: Explore device encryption features

Lesson 3: Explore device encryption features

- Examine BitLocker
- Evaluate BitLocker and TPMs
- Recover a BitLocker-encrypted drive
- Assess Encrypting Files System

Examine BitLocker

BitLocker encrypts the data that is stored on the operating system and other volumes by:

- Providing offline data protection
- Protecting all data stored on the encrypted volume
- Verifying the integrity of early startup components and boot configuration data
- Ensuring integrity of the startup process

BitLocker To Go allows encryption of removable media such as USB thumb drives

Evaluate BitLocker and TPMs

- TPM mode:
  - Locks the normal startup process until a user optionally supplies a personal PIN and/or inserts a USB drive that contains a BitLocker startup key
  - Performs system-integrity verification on startup components
- Non-TPM mode:
  - Uses Group Policy to allow BitLocker to work without a TPM
  - Locks the startup process similar to TPM mode, but the BitLocker startup key must be stored on a USB drive
  - Alternatively use a password

Recover a BitLocker-encrypted drive

When a BitLocker-enabled computer starts:

- BitLocker checks the operating system for conditions that indicate a security risk
- If a condition is detected:
  - BitLocker enters recovery mode and keeps the system drive locked
  - The user must enter the correct recovery key to continue

The BitLocker recovery key:

- Is a 48-digit key that unlocks a system in recovery mode
- Is unique to a particular BitLocker encryption:
  - Can be stored in AD DS
  - If stored in AD DS, search for it by using either the drive label or the computer’s account

Assess Encrypting Files System

- Encryption of files based on user
- Part of NTFS
- More Complex to Implement
- Scenarios
  - Protecting files on shared computers
  - Protecting files from privileged users
  - Sharing encrypted files with specific users

Knowledge Check

Lesson 4: Explore connection security rules

Lesson 4: Explore connection security rules

- Explain IPSec
- Explain connection security rules
- Explore authentication options
- Monitor security policies and active connections

Explain IPSec

- Is a suite of protocols that allows secure, encrypted communication between two computers over a unsecured network
- Has two goals: packet encryption and mutual authentication between systems
- Enables sending and receiving computers to send secured data to each other
- Secures network traffic by using encryption and data signing
- Uses policies to define the type of traffic that IPsec examines, how that traffic is secured and encrypted, and how IPsec peers are authenticated

Explain connection security rules

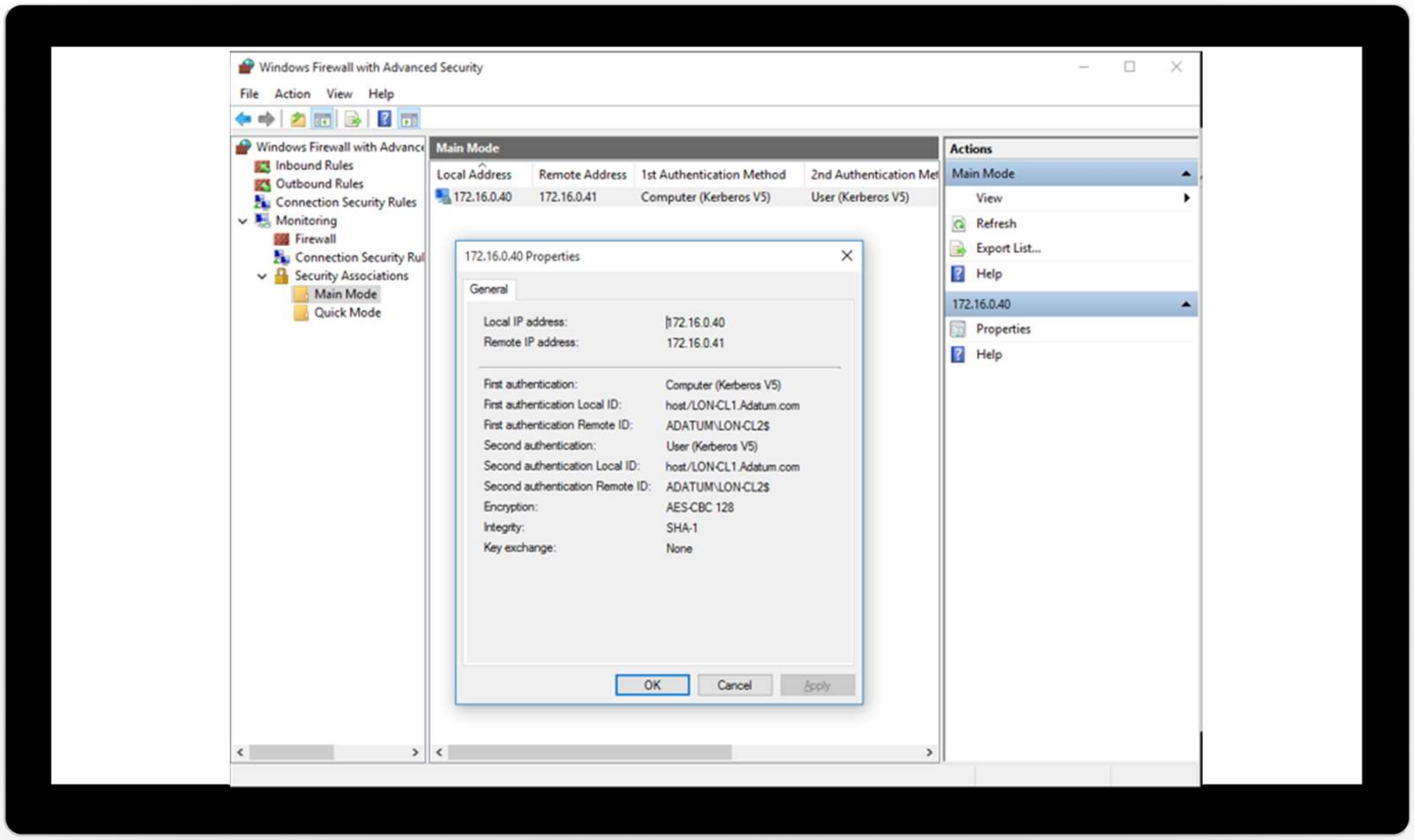
- Connection security rules involve:
  - Authenticating two computers before they begin communications
  - Securing information that is sent between two computers
  - Using key exchange, authentication, data integrity, and data encryption (optionally)
- How firewall rules and connection rules are related:
  - Firewall rules allow traffic through, but do not secure that traffic
  - Connection security rules can secure the traffic, but depend on a firewall rule to allow traffic through the firewall

Explore authentication options

When using the New Connection Security Rule Wizard to create a new rule, you use the Requirements page to choose one of the following:

Option	Description
Request authentication for inbound and outbound connections	Ask that all inbound/outbound traffic be authenticated, but allow the connection if authentication fails
Require authentication for inbound connections and request authentication for outbound connections	• Require that inbound traffic be authenticated, or it will be blocked • Outbound traffic can be authenticated, but will be allowed if authentication fails
Require authentication for inbound and outbound connections	Require that all inbound/outbound traffic be authenticated, or the traffic will be blocked

Monitor security policies and active connections



Knowledge Check

Lesson 5: Explore advanced protection methods

Lesson 5: Explore advanced protection methods

- Explore the Security Compliance Toolkit
- Use Applocker to control applications
- Use AppLocker to control Universal Windows Platform apps
- Secure data in the enterprise
- Understand Microsoft Defender for Endpoint

Explore the Security Compliance Toolkit

The key features of Security Compliance Toolkit include:

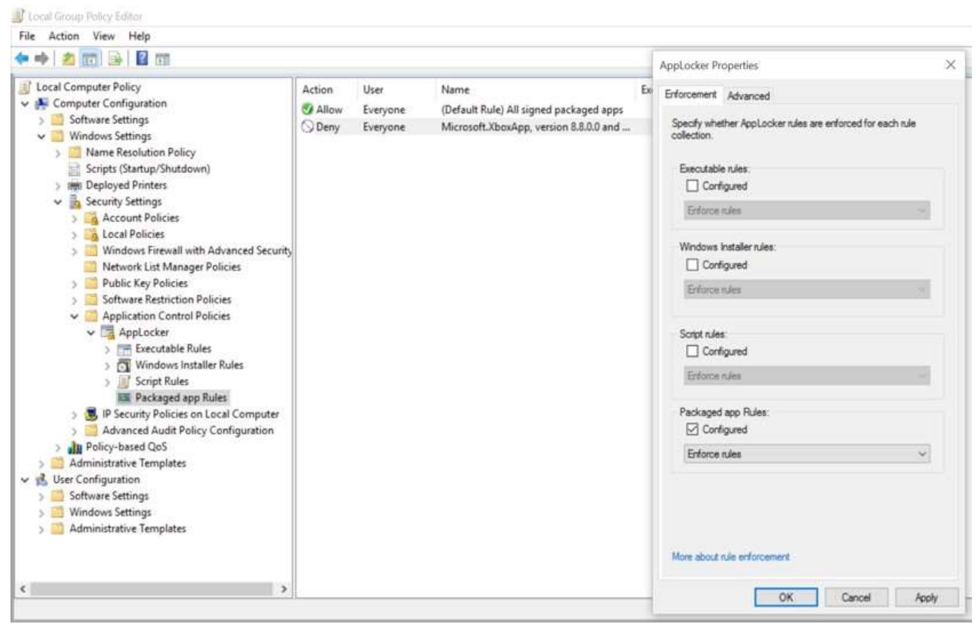
- Security baseline management features to manage the security and compliance process efficiently
- Baselines that are based on Microsoft security guide recommendations and industry best practices
- Compare GPOs with current settings using the Policy Analyzer tool

Use Applocker to control applications

- AppLocker is a security feature that enables you to specify exactly what applications are allowed to run on user PCs and devices
- Benefits of AppLocker:
  - Controls how users can access and run applications
  - Ensures that user PCs and devices are running only approved, licensed software

Use AppLocker to control Universal Windows Platform apps

- Configurable in Group Policy



Secure data in the enterprise

- Windows Device Health Attestation
- Windows Information Protection
- VPN Profiles

Understand Microsoft Defender for Endpoint

- Windows Defender Application Control
- Microsoft Defender Device Guard
- Microsoft Defender Credential Guard
- Microsoft Defender Application Guard
- Microsoft Defender Exploit Guard

Knowledge Check

Practice Labs

Configuring Microsoft Defender Antivirus and Windows Security

Configuring Firewall and Connection Security

Configuring and Troubleshooting BitLocker

Module Review

Microsoft 365